



Self-renewing zwitterionic silicone hybrid coatings with multi-modal bacterial resistance for superior anti-biofouling performance

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ABSTRACT

Zwitterion-modified silicone coating has gained prominence as eco-friendly antifouling materials due to the inherent resistance to microbial adhesion through hydrophilic charge rich feature. However, zwitterion-modified silicone exhibits poor static antibacterial efficacy as bacterial accumulation gradually compromises their anti-fouling capability. To address this issue, we innovatively propose a dynamic hybrid coating architecture integrating flexible polydimethylsiloxane (PDMS) chain segments, zwitterionic sulfobetaine silane (SBSi) with cinnamaldehyde (CAL) and terephthalaldehyde (TPA) via Schiff-base linkages (-C=N-) formed through dehydration condensation. This unique architecture retains the bacterium-repellent properties and active antibacterial functions characteristic of zwitterionic polymers. Crucially, the reversible hydrolysis of Schiff-base bonds enables sustained release of antimicrobial aldehyde derivatives. This promotes continuous self-renewal of fouled interfaces, facilitates the release of dead bacteria from the surface, and effectively enhances sustained static antimicrobial performance. The optimized coating achieved a sustained bactericidal rate exceeding 99.9 % over a 7-day period, representing an 81 % improvement over single polysiloxane coatings and a 36.5 % improvement over traditional zwitterionic silicone systems. In addition, the Schiff-base crosslinked zwitterionic silicone hybrid polymer undergoes environmentally benign degradation into small units, effectively circumventing the microplastic generation. This work establishes a multi-action antifouling strategy that combines passive surface resistance with dynamic antimicrobial activity while ensuring ecological compatibility, offering new insights for developing sustainable marine coatings.

1. Introduction

The threat of marine biofouling, the attachment of microorganisms, algae, and invertebrates to underwater surfaces [1], severely limits the efficient and long-term service life of marine infrastructure [2], which increased fuel consumption [3], sensor inaccuracies [4], and accelerated metal corrosion [5]. Biological fouling encompasses a hierarchical process initiating from the adsorption of proteins and polysaccharides, progressing through the colonization of microorganisms (e.g., bacteria and microalgae) to form biofilms, and ultimately establishing complex ecological communities dominated by macro-organisms such as barnacles and macroalgae [6]. The prevention of microbial adhesion (particularly bacterial primary attachment) and the inhibition of biofilm formation have been recognized as critical priorities in antifouling

strategies. Conventional antifouling materials achieve broad-spectrum bactericidal efficacy through controlled release of active agents such as cuprous ions and organotin compounds, effectively disrupting biofilm formation mechanisms. However, these materials are increasingly restricted due to environmental concerns associated with the discharge of highly toxic compounds [7]. Current research prioritizes the replacement of toxic traditional systems, such as organotin-based stripping coatings, with environmentally friendly coatings [7].

Organic silicone materials, typified by polydimethylsiloxane (PDMS), possess low surface energy and excellent environmental compatibility. Their inherent hydrophobicity and highly flexible Si-O-Si backbone structure significantly reduce microbial adhesion strength, typically only requiring hydrodynamic shear forces to achieve optimal antifouling performance. However, under static conditions, these

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materials remain susceptible to microbial adhesion, limiting their practical utility application [8]. Recent research has therefore focused on combining hydrophobic and hydrophilic components by grafting hydrophilic polymer, such as polyethylene glycol (PEG) or zwitterionic polymers [9]. This approach aims to integrate the fouling-release properties of silicones with the fouling-resistant properties of hydrogels, creating organosilicon-hydrogel hybrid systems to enhance the static biofouling resistance of PDMS-based coatings. Among these modifiers, zwitterionic polymers, a new generation of eco-friendly antifouling systems, have attracted significant attention due to inherent fouling resistance and contact-killing capabilities [10]. Each repeating unit of zwitterionic polymers contains both cationic and anionic groups, maintaining overall charge neutrality. Through electrostatic interactions and hydrogen bonding with water molecules, they form dense hydration layers at interfaces that effectively inhibit protein adsorption and microbial attachment [11]. Furthermore, the positively charged moieties (e.g., quaternary ammonium groups) of zwitterions can bind to negatively charged bacterial membranes, disrupting cellular integrity and inducing leakage of intracellular contents [12]. For instance, Chen et al. [13] developed a hyperbranched zwitterionic-siloxane coating, in which the zwitterionic components migrate to the coating surface in a humid environment to form a hydrated layer, achieving approximately 93.5 % antibacterial efficiency. Nevertheless, under long-term static service conditions, these surfaces become readily covered by inorganic deposits and dead bacterial cells, leading to the eventual loss of antifouling performance [14].

Maintaining persistent antifouling functionality requires continuous detachment of inactivated microorganisms from the fouled interface, ensuring sustained exposure of the zwitterionic hydration layer at the surface. However, achieving such dynamic stability remains a significant challenge. Antifouling coatings based on degradable polymers, particularly those utilizing reversible covalent bonds, have emerged as promising solutions for creating eco-friendly low fouling surfaces with dynamic stability. Their antifouling mechanisms involve surface degradation, where dissolved degradation products continuously shed into seawater, forming self-renewing interfaces that induce passive detachment of fouling organisms. Recent advancements highlight poly-Schiff base coatings incorporating dynamic imine bonds ($C=N$) as a breakthrough innovation in antifouling materials, which inherently resist bacterial and microbial adhesion through their degradable nature and surface dynamism [15]. The reversible imide covalent bonds in the poly-Schiff base group chain can spontaneously degrade in an aqueous environment into aldehydes, such as cinnamaldehyde (CAL) and amine-based small molecules with antibacterial effects, effectively inactivating surface bacteria [16,17]. This natural degradation process of Schiff base chains prevents microplastic pollution while maintaining exceptional antifouling efficacy under static conditions, independent of hydrodynamic forces. Nevertheless, the intrinsic antifouling performance of Schiff base polymer remains limited, necessitating synergistic enhancement strategies, which include integration with antifouling agents, photocatalysts antibacterial agents [18], polyphosphates, and chitosan [8,19] to amplify their functional scope.

Previous work has developed degradable organosilicon hydrogel hybrid coatings [20]. However, because polyethylene glycol lacks inherent antibacterial properties, microorganisms may proliferate upon contacting the coating, forming stubborn biofilms. Although zwitterion-modified silicone coatings combine fouling resistance, contact killing, and conditional fouling-release capabilities, they remain constrained by persistent challenges in static fouling scenarios and long-term durability. The degradable poly-Schiff base coating exhibits superior dynamic surface stabilization properties, and the hydrolysis derived product CAL demonstrates eco-friendly antimicrobial functionality. However, integrating these complementary features into a unified silicone hydrogel polymer architecture demonstrates immense application potential, while presenting significant technical challenges. For instance, the strong hydrophilicity of zwitterionic components may lead

to poor dispersion within the PDMS matrix [10]. To address these challenges, our designed hybrid coating system utilizes molecular engineering to sequentially introduce zwitterionic short-chain sulfobetaine silane (SBSi), terephthalaldehyde (TPA) crosslinker, and the natural antibacterial agent CAL into the amino-functionalized PDMS side chains. This constructs a multifunctional platform integrating flexible PDMS segments, contact-killing zwitterionic domains, dynamic covalent crosslinking nodes, and intelligently releasable natural antibacterial agents. The system establishes a crosslinked network through the formation of dynamic imine bonds between the difunctional TPA and the zwitterion-modified PDMS chains. This architecture enables the sustained release of TPA and CAL while exposing the zwitterionic layer during service. Simultaneously, structural integrity is maintained through dynamic bond reconfiguration, thereby creating a self-renewing dynamic antibacterial surface. This hybrid coating synergizes static antifouling mechanisms (the zwitterion-induced formation of a hydration layer and an antibacterial layer) with dynamic surface reconfiguration (coating degradation and antibacterial agent release enabled by imine reversibility) to achieve outstanding biofouling inhibition and extended service life. This approach is expected to provide both theoretical insights and practical solutions for marine and biomedical applications.

2. Experimental

2.1. Materials

3-Aminopropyltrimethoxysilane (APMDS), 3-aminopropyltriethoxysilane (KH550), 3-glycidyloxypropyltrimethoxysilane (KH560), (*N,N*-dimethyl-3-aminopropyl)trimethoxysilane (DMASi), terephthalaldehyde (TPA), cinnamaldehyde (CAL) and 1,3-propanesultone were obtained from Aladdin. 3-Aminopropyltriethoxysilane (KH550) and dibutyltin dilaurate (DBTDL) were purchased from Macklin Co. Ltd. Hydroxyl-terminated polydimethylsiloxane (OH-PDMS, $M_n = 400$ g/mol) was supplied by Gelest. Phosphate-buffered saline (PBS) was acquired from Beijing Solarbio Science & Technology Co., Ltd. *Escherichia coli* (E3342B) was procured from Shanghai Luwei Technology Co., Ltd. All chemicals were of analytical grade and used without further purification.

2.2. Synthesis of zwitterionic sulfobetaine (SBSi) monomer

Add 0.61 g of 1,3-propanesultone to 6.59 g of solute ethanol and shake thoroughly until dissolved. Once no visible particles remain in the ethanol solution, add 1.04 g of DMASi (Fig. 1a). Then, place the mixture under stirring at room temperature for 24 h to synthesize the sulfobetaine monomer.

2.3. Preparation of hybrid coating

Preparation of Polysiloxane Coating (Pdms): OH-PDMS (16.00 g), APMDS (8.16 g), KH550 (1.11 g), ethanol (6.00 g) and DBTDL (0.1 g, catalyst [21]) are combined in a reaction flask and stirred to form a homogeneous mixture. After dropwise addition of water (0.31 g) under stirring, the mixture was vacuum-mixed at room temperature for 1 h, then reacted at 80 °C under N_2 for 12 h to yield a transparent organosilicon prepolymer solution. The prepolymer was then mixed with KH560 (12.99 g) and stirred at room temperature for 1 h to initiate a ring-opening reaction, forming the polysiloxane polymer (Fig. S1) [13]. Approximately 130 μ L of the polymer was applied uniformly onto a glass substrate (coating area: 2.5 cm $\times$ 2.5 cm), air-dried to remove solvents, and thermally cured at 80 °C for 24 h. The final product was labeled as Pdms.

Preparation of Zwitterionic Organosilicon Coatings (5%a/15%a-pdms): The coating was formed by adding 5 % and 15 % sulfobetaine solutions with ethanol as the solute (5 %:1.03 g DMASi, 0.61 g 1,3-

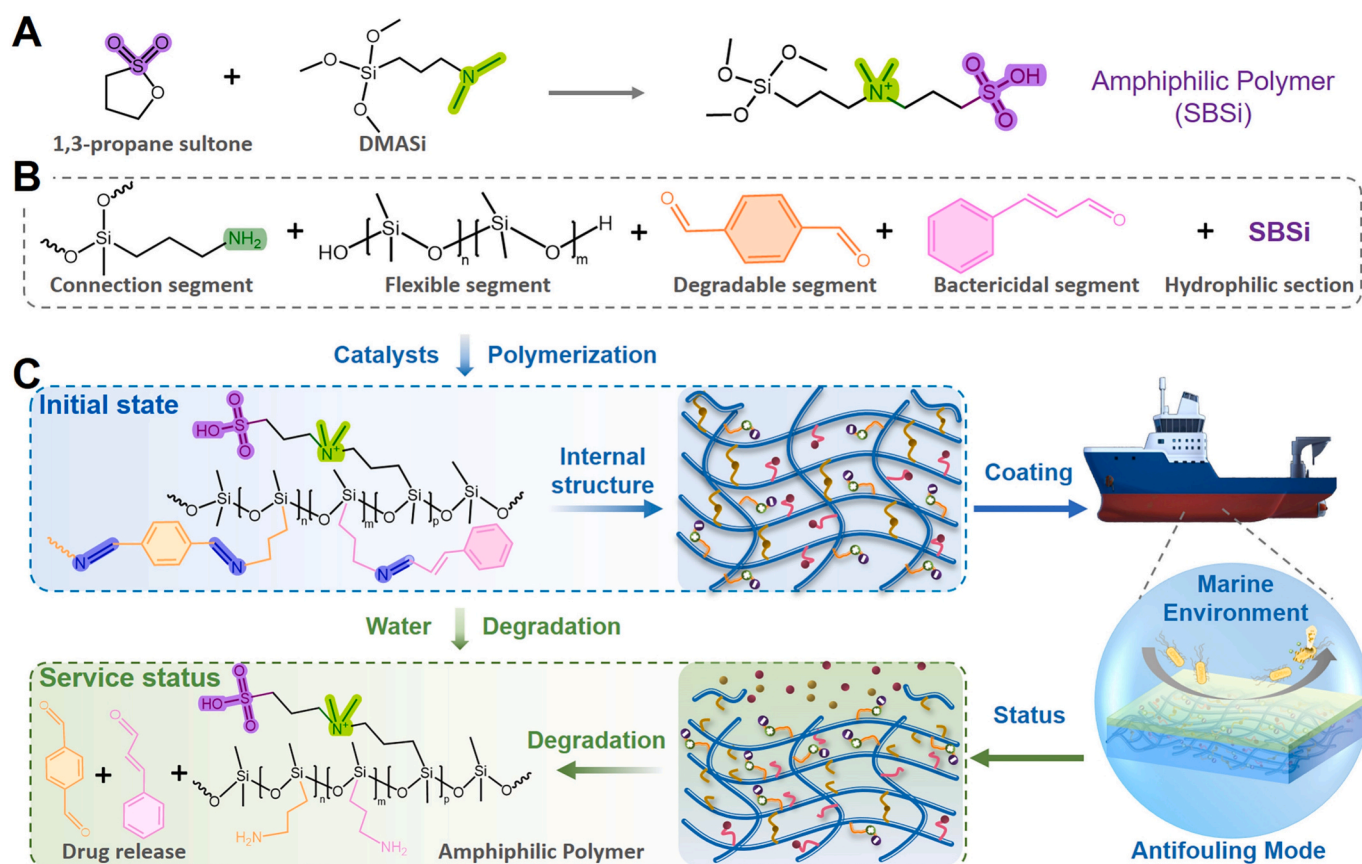


Fig. 1. Schematic illustrating the preparation and degradation process of the hybrid coatings. (a) Synthesis process of zwitterionic sulfobetaine silane (SBSi). SBSi is generated through the click reaction between propane sultone and amino groups. (b) Reaction forming a Schiff base-crosslinked zwitterionic organosilicon polymer. (c) Degradation and anti-fouling mode of Schiff Base Crosslinked Zwitterionic Polysiloxane coatings. When the coating is applied in a seawater environment, it degrades to release small molecule aldehydes and resists fouling.

propanesultone, and 6.59 g ethanol, with zwitterions constituting 5 % of total mass; 15 % solution proportionally scaled) in APDMS (8.16 g), KH550 (1.11 g) and OH-PDMS (16.00 g). Under stirring, 0.45 g water was added dropwise, and the mixture was heated at 80 °C under N₂ for 12 h to yield a transparent zwitterionic organosilicon solution. Later, KH560 (12.99 g) was added, and the mixture was stirred at room temperature for 1 h to preliminarily carry out the ring-opening reaction between epoxy groups and amino groups, as well as the condensation reaction among silicate esters, yielding a more viscous zwitterionic organosilicon polymer (Fig. S2). Approximately 150 µL of the polymer was applied uniformly onto a glass substrate (coating area: 2.5 cm × 2.5 cm), air-dried to remove solvents, and thermally cured at 80 °C for 24 h. The formed coatings were designated 5%a-pdms and 15%a-pdms.

Preparation of Schiff Base Crosslinked Zwitterionic Polysiloxane Coatings (Sb-pdms/5%a-sb-p/15%a-sb-p): Used APDMS (8.16 g), KH550 (1.11 g) and OH-PDMS (16.00 g) were mixed with 2.68 g of TPA, 1.98 g of CAL, and 5 % or 15 % zwitterion content. The mixture is stirred under N₂ at room temperature for 1 h, then heated to 80 °C for 4 h to produce a transparent pale-yellow solution. The formed coating was named 5%a-sb-p and 15%a-sb-p, respectively (Fig. 1b). A control coating, Sb-pdms, was synthesized using the same method without zwitterions. All three coatings are yellowish and transparent.

2.4. Characterization and testing

2.4.1. Characterization and testing

The Pdms, 5%a-pdms, 15%a-pdms, Sb-pdms, 5%a-sb-p and 15%a-sb-p were dissolved in CDCl₃ and analyzed by ¹H NMR spectroscopy (400 MHz AVANCE III) to determine molecular structures. FTIR (Nicolet

6700) and confocal Raman spectroscopy were used to record absorption peaks in the ranges of 400–4000 cm⁻¹ and 300–1800 cm⁻¹, respectively, monitoring chemical changes in the coatings before and after immersion. Coating transmittance was tested via UV-Vis-NIR spectrophotometer (LAMBDA) using quartz slides as substrates. Thermal properties were analyzed using differential scanning calorimetry (DSC; NETZSCH DSC 214) over a temperature range of –150 to 150 °C at a heating rate of 10 °C/min (sample mass: 10 mg each), and thermogravimetric analysis (TGA; NETZSCH TG209F1) spanning 30–800 °C at a heating rate of 10 °C/min (sample mass: 5 mg each) with derivative thermogravimetry (DTG). Surface wettability was monitored dynamically using a contact angle analyzer (DCAT 21). Degradation kinetics were analyzed using UV-vis-NIR (Lambda 950) to track absorption peaks over time. The coating morphology and bacterial adhesion were observed by SEM (Zeiss Sigma 300) at 3 kV, while a colony counter (SCAN 300) quantified bacterial survival (CFU/mL) on coatings and in solution to assess antibacterial efficacy.

2.4.2. Degradation test

To facilitate detection and elimination of interference from unknown ions, coating samples (Pdms, 5%a-pdms, 15%a-pdms, Sb-pdms, 5%a-sb-p, and 15%a-sb-p) with a size of 2.5 cm × 2.5 cm were completely immersed in 25 mL of ultrapure water. The changes in absorbance of the different samples over degradation days were recorded using a UV spectrophotometer. Two parallel groups were established to compare degradation under static and dynamic conditions: one group in a 20 °C shaking incubator (simulating flow) and the other in a static 20 °C incubator. Absorption intensity (200–900 nm) was measured every 24 h. Degradation products were structurally identified via characteristic

absorption peaks (e.g., $\pi \rightarrow \pi^*$ transition peak of benzaldehyde at 262 nm). Solution concentrations were calculated using peak intensity and standard calibration curves, and degradation reaction rates were quantified.

2.4.3. Antibacterial activity of hybrid coating

The antibacterial properties of the coatings (Pdms, 5%a-Pdms, 15% a-Pdms, Sb-pdms, 5%a-sb-p, 15%a-sb-p; 2.5 cm $\times$ 2.5 cm) were evaluated via quantitative colony counting. Fresh coatings and reference samples (blank glass, epoxy resin EP, polystyrene PS, pure PDMS) were immersed in centrifuge tubes with 25 mL ultrapure water for 24 h, then replaced with 25 mL bacterial suspension (10^8 CFU/mL) and incubated at 37 °C (120 rpm) for 24 h to ensure full contact with coatings. After removal, residual bacteria on the coating surfaces were absorbed using sterile nonwoven fabric. The samples (Pdms, 5%a-Pdms, 15%a-Pdms, Sb-pdms, 5%a-sb-p, 15%a-sb-p) were transferred to fresh PBS (25 mL) and statically incubated at 20 °C for 24 h. Antibacterial data were collected from two aspects: (1) Antibacterial activity in immersion solution: The solution was vortexed for 2 min, and 100 μ L was spread on NA agar plates (10 cm diameter). Colonies were counted after 18 h incubation at 37 °C using a colony counter. (2) Surface antibacterial activity: Samples were transferred to fresh PBS (5 mL), vortexed for 2 min to disperse surface bacteria, and plated similarly for counting. Pdms coatings and blank glass served as control and reference, respectively. Each experiment was independently repeated three times.

Antibacterial durability of composite coatings: They Samples (Pdms, 5%a-pdms, 15%a-pdms, Sb-pdms, 5%a-sb-p, and 15%a-sb-p and blank glass) were then placed in 25 mL of bacterial suspension (10^8 CFU/mL) and incubated at 37 °C with shaking (120 rpm). Samples were retrieved at 1, 3, 6, 12, and 24 h. After vortexing the tubes vigorously for 2 min, 100 μ L of the solution was spread on NA agar plates and incubated at 37 °C for 18 h. Each test was independently repeated three times, with Pdms as the control and blank glass as the reference. The antibacterial rate (%) was calculated as:

$$A(\%) = \frac{N_0 - N_a}{N_0} \times 100\%$$

In this study, N_0 is defined as the number of colonies counted on the pure glass surface or in the control group bacterial solution, while N_a represents the number of colonies obtained from counting on the prepared coating surface or in the coating immersion suspension [18].

2.4.4. Morphological analysis of bacteria

The bacterial morphology on the coating surface and in the degraded bacterial solution was observed using SEM. Samples immersed in 25 mL bacterial suspension (10^8 CFU/mL) for 24 h were removed, washed twice with ultrapure water (5 min each), and then fixed with 2.5 % glutaraldehyde for 20 min. Finally, the samples were rinsed twice with ultrapure water (5 min each) to remove residual fixative and then air-dried. Meanwhile, the bacterial solution from the centrifuge tube was dropped onto a clean glass slide, dried at 40 °C and fixed using the same protocol. All samples were sputter-coated with gold for SEM imaging.

3. Results and discussion

3.1. Structure and morphology characterization

The chemical structures of polysiloxane (Pdms), zwitterionic organosilicon polymer (A-pdms) (A = 5%a and 15%a), and Schiff base polymer (Sb-pdms) were analyzed by ^1H NMR spectroscopy to investigate the synergistic effects between zwitterionic groups and Schiff base moieties on the organosilicon framework. (Fig. 2a). Characteristic peaks at 3.7 ppm and 2.6 ppm in the siloxane polymer were assigned to the secondary amine (-NH-) on the aminopropyl group and the methyl (-CH₃) on APDMS, respectively. In the zwitterionic polymer (SBSi), peaks at 1.4 ppm, 2.6 ppm, and 3.7 ppm corresponded to the methyl (-CH₃) and methylene (-CH₂-) groups linked to the quaternary ammonium moiety [12,13,22,23]. Notably, a new peak at $\delta = 8.28\text{--}8.30$ ppm (labeled as f) emerged in Sb-pdms, corresponding to the conjugated imine group (-C=N-) of aromatic rings [18]. The characteristic resonances conclusively validate the successful synthesis of the target silane polymer.

The chemical structures of samples Pdms, A-pdms, Sb-pdms, and A-sb-p before and after immersion were characterized using FT-IR

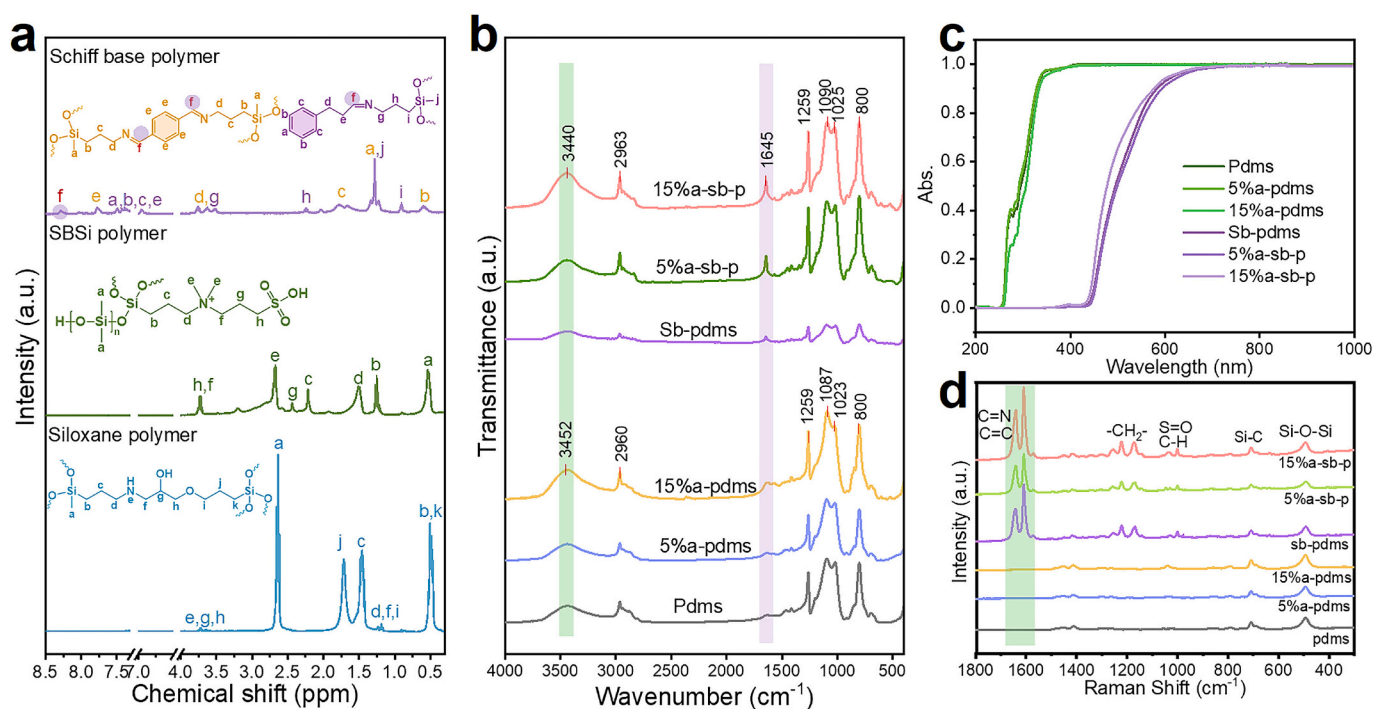


Fig. 2. Coating structure characterization. (a) ^1H NMR spectra of Siloxane polymer (Pdms), zwitterionic organosilicon (SBSi polymer), and Schiff base polymer Sb-pdms. (b) FTIR spectra. (c) UV transmittance. (d) Raman spectra of silicone hybrid coatings.

spectroscopy (Fig. 2b). (Before immersion: untreated; after immersion: coatings were completely soaked in ultrapure water for 7 days). The absorption band in the $3300\text{--}3700\text{ cm}^{-1}$ range corresponds to the typical stretching modes of carbon-nitrogen heterocyclic compounds and the stretching vibrations of adsorbed O—H and residual N—H [24]. The peak at 2960 cm^{-1} is attributed to the C—H stretching vibration of alkyl groups [25]. Two strong absorption peaks at $1025\text{--}1090\text{ cm}^{-1}$ and peaks at 800 cm^{-1} and 1259 cm^{-1} were assigned to Si—O—Si stretching, Si—C bending vibration, and Si—C stretching vibration in PDMS segments, respectively [26]. The absorption in the $1600\text{--}1700\text{ cm}^{-1}$ range corresponded to C=N stretching (Schiff base), C=C aromatic bonds, and S=O stretching in sulfobetaine silane [25,27], which was consistent with the structural features of zwitterionic and poly-Schiff base polymers. After immersion, the stretching vibrations of the N—H peaks (3400 cm^{-1}) of the six samples were significantly enhanced. The peaks at 1600 cm^{-1} of A-pdms and Sb-pdms were enhanced (Fig. S3). For Pdms and A-pdms, it is due to the aggregation of hydrophilic N—H/S=O on the coating surface after encountering water, resulting in enhanced vibrations [28]. For the Schiff base polymers (Sb-pdms and A-sb-p), immersion in water leads to the reversible hydrolysis of the Schiff base to form amino and aldehyde groups C=O (1600 cm^{-1}) [29], resulting in

enhanced peaks. Meanwhile, the peak positions hardly change, indicating the stability and integrity of the structure.

The UV–Vis absorption spectra of different samples were characterized using quartz slides as substrates. The absorption peaks of Pdms and A-pdms samples are located around 250 nm . Upon incorporating Schiff base, Sb-pdms and A-sb-p exhibited a significant red shift (maximum absorption wavelength $\sim 430\text{ nm}$) (Fig. 2c) due to $\pi \rightarrow \pi^*$ transitions from increased conjugated double bonds, leading to a reduced energy gap [29]. Notably, all samples ($\sim 500\text{ }\mu\text{m}$ thick) showed an average transmittance of $\geq 99.5\%$ in the $700\text{--}1000\text{ nm}$ range, indicating excellent optical clarity. The Raman spectra of the synthesized samples was recorded using a 785 nm laser in the $300\text{--}1800\text{ cm}^{-1}$ range (Fig. 2d). The bending vibration of $-\text{CH}_2-$ in the epoxy group [30] appeared at $1450\text{--}1470\text{ cm}^{-1}$, while the bending vibration of Si—O—Si was observed in the $400\text{--}500\text{ cm}^{-1}$ region [31]. The stretching vibration of the PDMS's Si—C bond was detected at $750\text{--}800\text{ cm}^{-1}$. Vibrations associated with the imine and phenyl rings were identified in the $1180\text{--}1260\text{ cm}^{-1}$ region [32]. The sulfonic acid group ($-\text{SO}_3\text{H}$) in sulfobetaine silane and C—H vibrations in the benzene ring were observed at $1000\text{--}1060\text{ cm}^{-1}$ [33]. After grafting the Schiff base side chains, the C=C stretching vibrations from CAL and TPA molecules, the C=N stretching vibrations in

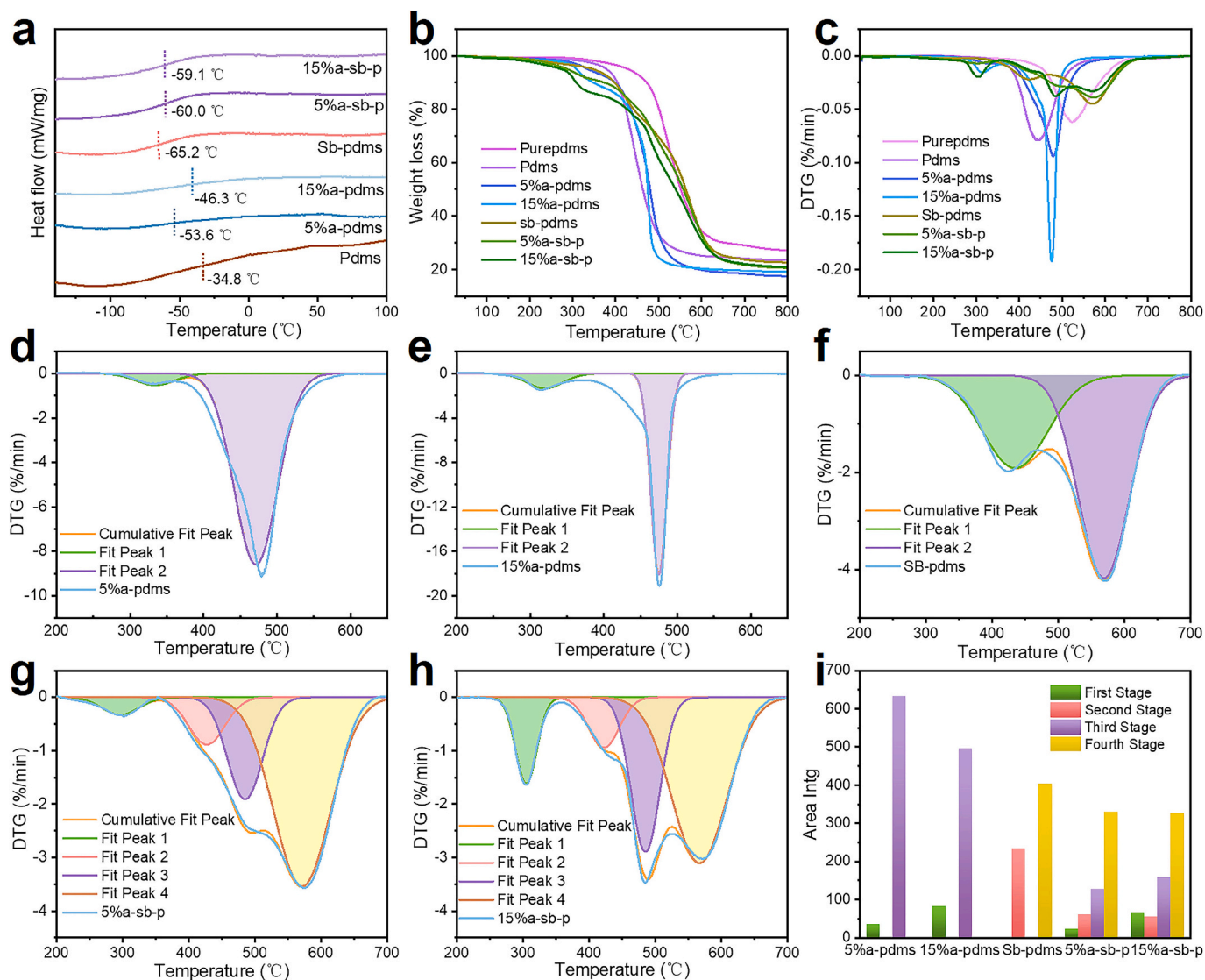


Fig. 3. Thermal properties of the hybrid coating. (a) The DSC curves of the samples. (b–c) present the TG and DTG curves of pure PDMS, pdms, A-pdms, Sb-pdms and A-sb-p, respectively. (d–h) display the DTG peak deconvolution graphs for 5%a-pdms, 15%a-pdms, Sb-pdms, 5%a-sb-p and 15%a-sb-p. (i) The peak deconvolution of the DTG curves for different thermal decomposition stages, along with the integrated peak areas for each stage.

the Schiff base were all detected in the 1600–1650 cm^{-1} region [34] which confirmed the formation of Schiff base. After immersion, the $\text{C}=\text{N}$ bonds undergo reversible hydrolysis into $-\text{NH}_2$ groups and aldehyde monomers containing $\text{C}=\text{C}$ bonds (CAL and TPA). The dissolution of these aldehyde compounds in water leads to a decrease in peak intensity at 1600–1650 cm^{-1} [27] (Fig. S3). These results are consistent with the FT-IR analysis.

The thermal stability of the samples was investigated using differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA). The glass transition temperature (T_g), differential thermogravimetric (DTG), and thermogravimetric (TG) data were obtained under nitrogen (N_2) atmosphere. The T_g of the samples, including Pdms, 5%a-pdms, 15%a-pdms, Sb-pdms, 5%a-sb-pdms, and 15%a-sb-pdms, were measured by differential scanning calorimetry, yielding values of -34.8°C , -53.6°C , -46.3°C , -65.2°C , -60.0°C , and -59.1°C , respectively (Fig. 3a). Compared to Pdms, the T_g of zwitterion modified polydimethylsiloxane (A-pdms) decreased by 18.8°C (5 % content) and 11.5°C (15 % content), attributed to that zwitterionic groups disrupts the ordered arrangement of polydimethylsiloxane (PDMS), thus expanding the free volume of materials and thereby reducing its T_g . However, the higher T_g observed in 15%a-pdms (-46.3°C) compared to 5%a-pdms (-53.6°C) arises from electrostatic interactions between zwitterions, which restrict the mobility of molecular chain segments [35]. The subsequent introduction of Schiff base bonds ($\text{C}=\text{N}-$) into the side chains (sample A-sb-p), due to their reversibility, further reduces T_g and enhances the flexibility of the molecular chain, increasing the local mobility of the main chain at low temperatures [20].

The TG and DTG data provide further evidence of the thermal distinctions between the as-prepared materials. The 5 % weight loss temperatures ($T_5\%$) for 5%a-pdms, 15%a-pdms, 5%a-sb-p, and 15%a-sb-p were 335.6°C , 309.1°C , 287.4°C , and 274.1°C , respectively, showing a decreasing trend (Fig. 3b). These $T_5\%$ values were notably lower than those of Purepdms (429.2°C), Pdms (379.1°C), and Sb-pdms (347.0°C). The reduced $T_5\%$ for A-pdms and A-sb-p is primarily attributed to the thermal decomposition of zwitterionic sulfobetaine, with higher zwitterion content leading to lower $T_5\%$ values [36]. Additionally, the further decrease in $T_5\%$ for A-sb-p compared to A-pdms is due to the thermal decomposition of the Schiff base grafted onto the side chains. It is noteworthy that the $T_5\%$ values of all samples are above 270°C , indicating that the zwitterionic/Schiff base polymers possess good thermal stability. At 800°C , the residual carbon contents (C_{800}) of 5%a-pdms, 15%a-pdms, 5%a-sb-p, and 15%a-sb-p are 18.38 %, 20.26 %, 21.39 %, and 22.29 %, respectively. The higher C_{800} of the 15 % zwitterionic content samples compared to the 5 % samples is attributed to the formation of a dense ionic crosslinked network through strong electrostatic interactions between more positively and negatively charged groups, which inhibits the thermal motion of molecular chains and delays main-chain scission [37]. The higher C_{800} of A-sb-p relative to A-pdms is due to the incorporation of Schiff base. The benzene ring structures in CAL and TPA contribute to enhanced thermal stability. Due to TPA and siloxane produced a more tightly crosslinked structure, which suppresses segmental motion, reduces the release of small molecular fragments, and thereby retains more residual carbon. Furthermore, the zwitterion/Schiff base crosslinking on PDMS side chains creates a denser network, facilitating a more stable char layer that shields the internal structure from decomposition and improving coating stability [38].

The DTG curves describe the relationship between the mass change rate of the coating and temperature (Fig. 3c). The mass loss of the samples in the range of 200°C to 700°C is attributed to the decomposition of zwitterionic sulfobetaine silane (SBSi), Schiff base, and polysiloxane. According to the TG and DTG curves, the thermal decomposition of A-pdms is divided into two stages: the thermal decomposition of SBSi ($250\text{--}370^\circ\text{C}$) and the thermal decomposition of the PDMS chain ($370\text{--}600^\circ\text{C}$) [39] (Fig. 3d-e). During the zwitterion decomposition stage, 5%a-pdms and 15%a-pdms result in mass losses of

5.2 % and 14.2 %, respectively, with a DTG integral area ratio of approximately 1:3 (Fig. S4). The thermal decomposition of the PDMS chain results in mass losses of 94.8 % and 85.8 %, respectively, which is consistent with the content ratio of pdms in the samples. For Sb-pdms, there are two stages of thermal decomposition: the thermal decomposition of Schiff base ($300\text{--}460^\circ\text{C}$) and the PDMS chain ($460\text{--}700^\circ\text{C}$), resulting in mass losses of 36.8 % and 63.2 %, respectively (Fig. 3f). The thermal decomposition of A-sb-p is divided into four stages (Fig. 3g-h). The first stage ($250\text{--}350^\circ\text{C}$) and the third stage ($450\text{--}520^\circ\text{C}$) are due to the thermal degradation of SBSi. Unlike the first stage, the third stage also involves the thermal degradation of the Schiff base. In the first stage, 5%a-sb-p and 15%a-sb-p cause mass losses of 4.2 % and 10.8 %, respectively, matching the mass proportion of zwitterions in the samples. The second stage of the reaction ($350\text{--}450^\circ\text{C}$) is due to the thermal decomposition of the Schiff base [40], causing mass losses of 11.2 % and 9.0 %, respectively. In the third stage, the DTG integral area ratio is approximately 1:1, reflecting the combined decomposition of zwitterions and Schiff base (Fig. 3i). The fourth stage ($450\text{--}700^\circ\text{C}$) involves the thermal decomposition of the PDMS chain, causing mass losses of 61.2 % and 53.9 %, respectively (Table S1). The above conclusions are consistent with the theoretical solid content proportions. The thermal decomposition temperature of polydimethylsiloxane (PDMS) increased from 442°C to 475°C and further to 572°C with the incorporation of zwitterionic and Schiff base components, demonstrating enhanced thermal stability [37,38].

3.2. Degradation properties of coatings

In the Schiff base crosslinked zwitterionic organosilicon polymer coating, the imine bonds ($\text{C}=\text{N}$) undergo reversible degradation into aldehyde small molecules (CAL and TPA), avoiding marine microplastic pollution while forming a self-renewing surface to prevent microbial adhesion. CAL exhibits bactericidal properties [17], while TPA cross-links with siloxane side chains to form a tighter polymer network [23], thus enhancing the structural stability and showing significant potential in marine antifouling applications. This study systematically analyzed the chemical and physical property changes of the polymer coatings before and after degradation, focusing on morphological evolution, degradation kinetics, and antibacterial performance.

UV spectrophotometer testing was performed on the sample degradation solutions to obtain absorption spectra for analyzing degradation kinetics under varying immersion times and environmental conditions (the degradation behaviors of the coatings were evaluated in both ultrapure water and PBS solutions in this study, which exhibited similar degradation profiles (Figs. 4 and S5)). The aromatic aldehyde chromophores in the molecules facilitated $\pi\text{--}\pi^*$ electronic transitions in the $200\text{--}300\text{ nm}$ range, manifesting as the K-band [41]. The characteristic absorption peak at 253 nm (K-band) corresponded to the degradation product TPA, while the peak at 285 nm arose from CAL (Fig. S6). The conjugated double bonds in CAL conferred sensitivity of its electron cloud distribution to polar environments. The hydrophilicity of zwitterionic moieties enhanced surface hydration, creating highly polar microenvironments that promoted electronic transitions ($\pi\text{--}\pi^*$) in conjugated system of CAL, thereby accelerating degradation [42]. Consequently, higher zwitterionic content correlated with stronger CAL absorption peak intensity (Figs. 4a-c and S7). The residual coating mass ratio was quantitatively analyzed using a TPA calibration curve to obtain degradation rate curves (Fig. 4d-f). As immersion time increased, the degradation rate gradually decreased, with both static and flowing-state samples (e.g., FSB-pdms denotes flowing-state Sb-pdms) exhibiting stable degradation between 8 and 18 days. Based on TPA/CAL degradation rates and a $200\text{ }\mu\text{m}$ coating thickness (Table S2), the hybrid coating of Schiff base cross-linked zwitterionic organosilicon (15%a-sb-p) demonstrated a predicted service life of 9.6 years, while F15%a-sb-p showed 9.1 years. This long-term durability meets operational requirements for extended use in medical devices, marine infrastructure,

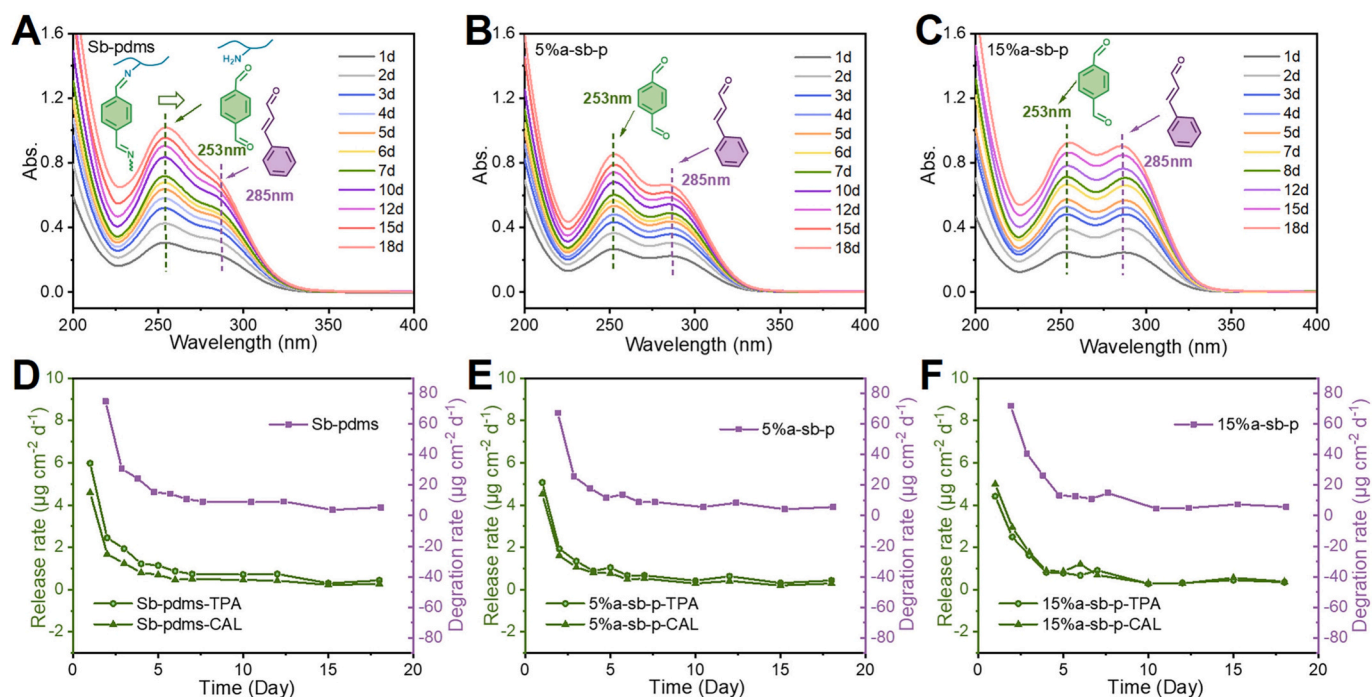


Fig. 4. (a-c) Show the UV absorption spectra of Sb-pdms and A-sb-p over different days of immersion in ultrapure water. (d-f) correspond to the coating degradation rates and the release rates of TPA and CAL.

submersible equipment, and related applications.

The surface morphology and elemental composition changes of Pdms, 15%a-pdms, Sb-pdms, and 15%a-sb-p before and after aqueous degradation were investigated (Fig. 5a-i). While all samples exhibited increases or decreases in elemental content after immersion, the surface distributions of C, N, Si, O, and S elements remained uniform.

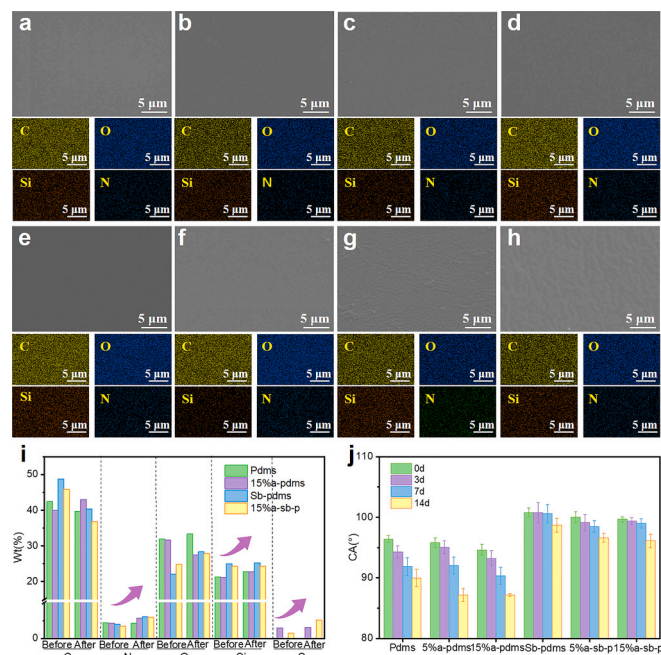


Fig. 5. (a-d) Surface morphology and elemental distribution (C, O, Si, N) of pdms, 15%a-pdms, Sb-pdms, and 15%a-sb-p. (e-h) Surface morphology and elemental distribution (C, O, Si, N) of pdms, 15%a-pdms, Sb-pdms, and 15%a-sb-p after immersion for 7 days. (i) Elemental content (C, N, O, Si, S) of samples before and after 7 days immersion in ultrapure water. (j) Contact angle measurements of pdms, A-pdms, Sb-pdms, and A-sb-p.

Specifically, for the siloxane polymers (Pdms and 15%a-pdms), Pdms showed negligible changes in elemental composition after water exposure. In contrast, 15%a-pdms, containing hydrophilic SO_3^- sulfobetaine siloxane segments, exhibited surface enrichment, leading to increased C, N, Si, and S content but a relative decreased in O. For the Schiff base polymers (Sb-pdms and 15%a-sb-p), the imine ($\text{C}=\text{N}$) groups underwent reversible hydrolysis in water, yielding a -NH_2 -containing Si-O-Si polymer chain while releasing CAL and TPA monomers (containing $\text{C}=\text{O}$ and $\text{-C}=\text{C}$). The dissolution of these aldehyde monomers into water resulted in elevated N, Si, and O content but reduced C. Despite these elemental variations, EDS mapping analysis confirmed that the coating maintained consistent elemental distribution profiles throughout the degradation process, further confirming its structural integrity and stability. In 15%a-pdms, surface pits (~ 10 nm in diameter) expanded to ~ 25 nm after 7-day immersion. This was attributed to microvoid formation caused by small molecule byproducts (e.g., alcohols) from the hydrolysis condensation of silane coupling agents. Incompatibility between sulfobetaine (an oligomeric surfactant) and the silane system exacerbated solvent gradient evaporation, creating surface tension driven pits or pores [43–45]. After immersion, the concave pores were enlarged due to swelling of the polymer network, which was caused by water adsorption through hydrogen bonding and ion-dipole interactions in zwitterionic polymers (Fig. S9) [11,46]. Notably, compared with 15%a-pdms, 15%a-sb-p exhibited no surface depressions owing to the self-renewing property of the surface Schiff base. Moreover, after soaking for 7 days, both Sb-pdms and 15%a-sb-p developed rough surfaces with uniform wavy patterns, demonstrating the occurrence of degradation.

Alterations in surface composition influence the resultant surface characteristics. To assess the surface properties between coating and water, we conducted systematic contact angle measurements following immersion in aqueous environments for varying durations (0, 3, 7, and 14 days) as illustrated in Fig. 5j. Comparative analysis revealed a distinct correlation between zwitterionic content and wettability: while the 5% a-pdms formulation exhibited a water contact angle of $96.0^\circ \pm 0.8^\circ$, this value decreased to $94.5^\circ \pm 0.9^\circ$ for the 15%a-pdms variant. This inverse relationship demonstrates that increased zwitterionic content enhances

surface hydrophilicity, a phenomenon attributable to the abundant zwitterionic groups (quaternary ammonium cations and sulfonate anions) present on the A-pdms surface. These amphiphilic moieties facilitate the formation of a hydration layer through strong dipole water interactions, thereby promoting complete surface wetting [47]. In contrast, the hybrid coatings crosslinked with Schiff base zwitterionic organosilicon networks achieved a contact angle of up to 99.6° (15%a-sb-p). Specifically, the contact angle of the hybrid coating (e.g., 15%a-sb-p) was 5.1 % higher than that of the zwitterionic organosilicon (e.g., 15%a-pdms) and 3.3 % higher than the unmodified organosilicon coating (Pdms). After 14 days of immersion, the contact angle of 15%a-pdms decreased by 7.8 % due to the hydrophilic nature of zwitterions, whereas 15%a-sb-p showed only a 3.5 % reduction. Moreover, during the 200-second dynamic contact angle measurement, a 5.9 % decrease was observed for 15%a-pdms, while 15%a-sb-p showed 4.4 % (Fig. S8). This is ascribed to the hydrophobic Schiff base structure (e.g., aromatic rings, imine groups) reducing surface polarity, combined with cross-linking between the Schiff base and zwitterionic organosilicon, forming a more compact hybrid coating that enhances hydrophobicity [48]. Remarkably, the hybrid coating A-sb-p maintained a contact angle $>96.1^\circ$ throughout the immersion period, demonstrating their hydrophobicity.

To further investigate elemental and chemical bonding changes induced by aqueous immersion, XPS analysis was conducted on Pdms, 15%a-pdms, Sb-pdms, and 15%a-sb-p before and after 7-day water exposure (Fig. 6a-b). XPS specifically probes the top 1–10 nm surface layer, unlike EDS (Energy Dispersive Spectrometer), which provides bulk elemental statistics. All samples showed C, N, O, and Si as primary constituents. Notably, S signals were additionally observed on the surfaces of zwitterionic sulfobetaine-containing samples (15%a-pdms and 15%a-sb-p) due to the sulfobetaine moieties (Fig. S10). Subsequent peak deconvolution further confirmed these chemical state alterations induced by immersion. (Figs. 6c-d and S11). The C 1s spectra of Pdms were deconvoluted into four peaks corresponding to C—Si (284.17 eV), C—C (284.8 eV), C—O (286.02 eV), and C—N (287.71 eV). After immersion, the hydrophilic -NH-[49] and -OH [50] groups generated by

the dehydration condensation of KH560 promoted surface accumulation of long carbon chains, increasing C/N signals and reducing O/Si, leading to elevated C—O and C—C bond contents in 7-Pdms. For 15%a-pdms, the C 1s spectrum was fitted with five peaks (C—Si: 284.22 eV, C—C: 284.8 eV, C—O: 285.84 eV, C—N: 286.45 eV, and C—SO₃⁻: 287.76 eV), confirming zwitterionic grafting. Due to the migration of sulfobetaine segments driven by hydrophilic zwitterions upon water exposure, C—C and C—O bonds significantly increased, with an upward trend also observed in C—N bonds. In the O 1s spectrum, it can also be observed that the peak value of the S=O bond increases after immersion (Fig. 6d), which enhanced the C/O/N/S signal levels while reducing Si content as the PDMS surface became covered. In the Schiff base-modified organosilicon Sb-pdms, hydrolysis of -C=N- bonds generated water-soluble aldehydes (-C=O) and hydrophilic amino groups (-NH₂), decreasing O content and increasing C/N signals. In contrast, in the Schiff base-modified zwitterionic organosilicon coating (15%a-sb-p), surface enrichment of zwitterions further elevated C/N/O/S and six deconvoluted peaks were observed (C—Si: 284.35 eV, C—C: 284.8 eV, C=C: 285.30 eV, C—N: 286.13 eV, C—SO₃⁻: 286.83 eV, and C=O/C=N: 288.13 eV), where C=C and C=N arose from the Schiff base structure. The presence of C—SO₃⁻, C=C, C=N, and C=O confirms successful grafting of sulfobetaine and Schiff base moieties onto PDMS [51–53]. This compositional evolution on the coating surface further corroborates that the degradation behavior and hydrophilic behavior promote the multifunctional response of the hybrid coating (forming a dynamic release and static antifouling surface).

3.3. Antibacterial properties of hybrid coatings

Escherichia coli (*E. coli*) is a commonly used model bacterium for investigating the antibacterial properties of coatings [9,10,16]. In this study, coating samples were incubated with bacterial suspensions (1×10^8 CFU/mL), and viable bacteria adherent to the surface or suspended in the solution were quantified by plate counting at different time points to calculate to determine antibacterial rates. The descriptive statistical methods were used to process these data. After 24 h of immersion in

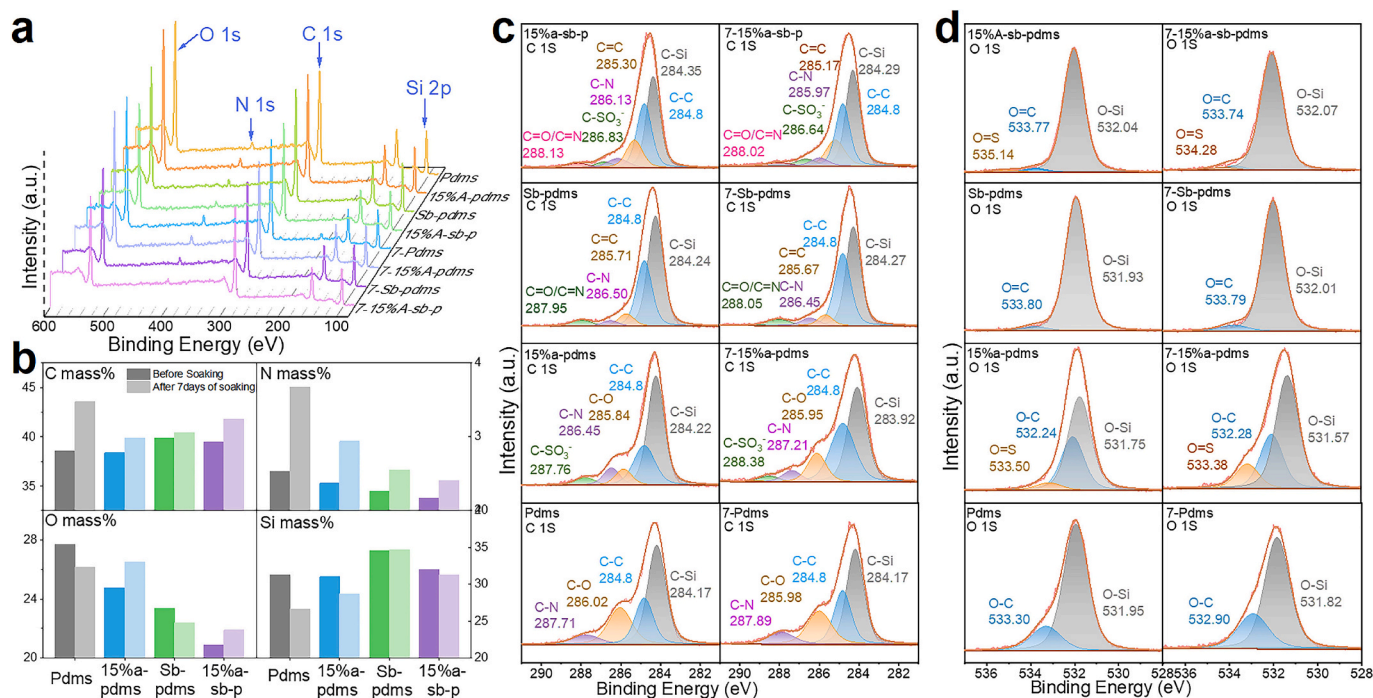


Fig. 6. (a) XPS survey spectra of polysiloxane Pdms, 15%a-pdms, Sb-pdms, 15%a-sb-p, and their counterparts after 7-day immersion (labeled as 7-Pdms, 7-15%a-pdms, 7-Sb-pdms, 7-15%a-sb-p). (b) Elemental content (C, O, Si, N) of Pdms, 15%a-pdms, Sb-pdms, and 15%a-sb-p before and after 7-day water immersion. (c) XPS spectra of C 1s and O 1s core levels for Pdms, 15%a-pdms, Sb-pdms and 15%a-sb-p before and after immersion in ultrapure water.

bacterial suspension, significant bacterial adhesion was observed on control samples (glass, epoxy resin (EP), polystyrene (PS), and pure PDMS). However, the zwitterion-modified organosilicon polymer (A-pdms) showed significantly reduced colony counts, while the Schiff base cross-linked zwitterionic organosilicon polymer (A-sb-p) surface and the immersed bacterial solution exhibited virtually no detectable colonies. These results demonstrate that the Schiff base cross-linked zwitterionic organosilicon polymer can effectively deactivate both surface-adhered and solution-phase bacteria (Fig. 7a, c and d). Due to the inherent hydrophobicity of PDMS derived from its low surface energy, bacterial adhesion remained relatively low. Upon crosslinking varying amounts of zwitterionic groups onto PDMS chains, bacterial colonization on the sample surfaces progressively increased with higher zwitterionic content. However, according to the bactericidal performance evaluation, the 15%a-pdms coating demonstrated a 90 % higher bactericidal rate compared to Pdms. This enhancement is attributed to the hydration layer formed by zwitterions upon water contact. While the hydration layer becomes covered by bacteria, the positively charged zwitterionic groups electrostatically interact with negatively charged bacterial surfaces, disrupting membrane integrity and consequently killing adhered bacteria on the coating surface (Fig. 7b-d). For the Schiff base hybrid

zwitterionic organosilicon polymer, almost no bacteria can be observed on the coating surface. The imine groups ($-C=N-$) enable continuous surface self-renewal to prevent bacterial colonization, avoiding the coverage of the hydration layer by bacteria and maintaining long term antibacterial efficacy [15]. Meanwhile, the controlled release of CAL can induce bacterial inactivation [17]. This multi-action mechanism of static resistance and dynamic release results in almost no bacterial adhesion on the coating surface and in the immersion solution, achieving an antibacterial rate of 99.9 % (Fig. 7d), demonstrates exceptional antibacterial performance.

Using SEM, the bacterial morphologies on the surfaces of Pdms, A-pdms, Sb-pdms, and A-sb-p were observed on the sample surfaces and in the bacterial suspensions after soaking in a bacterial solution of 10^8 CFU/mL for 24 h. On the surfaces and in the solutions of Pdms and A-pdms, rod shaped *Escherichia coli* with a length of approximately $1\ \mu\text{m}$ were detected, showing intact cytoplasm and cell membranes with no signs of damage. (Fig. 7b). The hydrophilic nature of zwitterions led to substantial bacterial adhesion on zwitterionic organosilicon coatings. In contrast, Sb-pdms and A-sb-p surfaces showed minimal bacterial adhesion, with adhered bacteria displaying abnormal curling. Bacterial suspensions of 5%a-pdms and 15%a-pdms exhibited cell membrane rupture

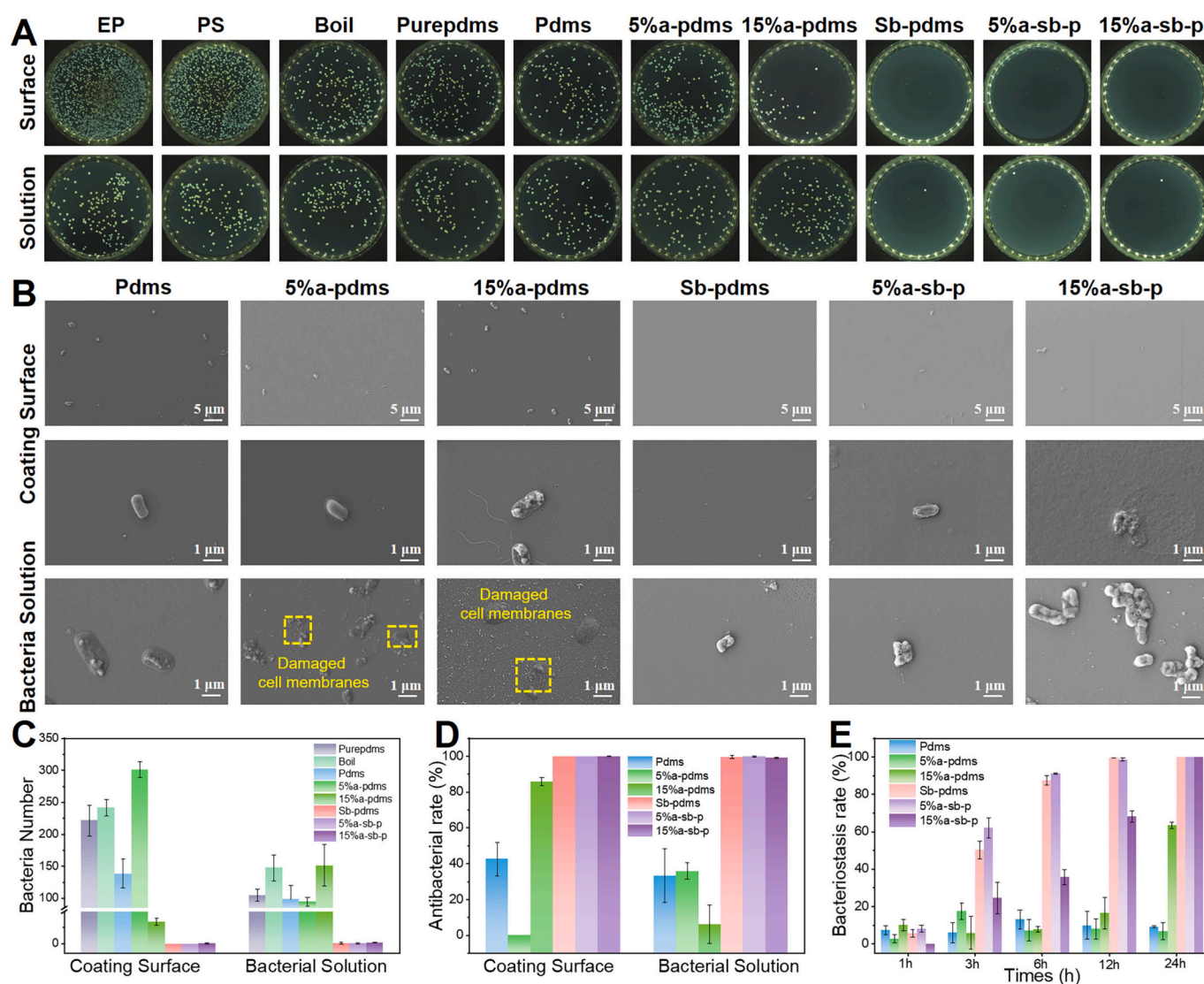


Fig. 7. (a) Viability of *Escherichia coli* on bare substrates versus glass-based coated samples, and in corresponding bacterial suspensions after material immersion. (b) Bacterial morphology observed via SEM on the sample surface and in the immersed bacterial solution. (c) Quantified bacterial count on surfaces and in the immersed bacterial solution. (d) Antibacterial rate. (e) Time-dependent antibacterial rate of the test sample at 1 h, 3 h, 6 h, 12 h, and 24 h.

and cytoplasmic leakage, along with leaked cellular debris such as damaged membranes and cytoplasmic components (Fig. S12). The positively charged zwitterions interacted with the negatively charged bacterial membranes, causing membrane rupture and cytoplasmic leakage [54]. Conversely, bacterial suspensions in Sb-pdms and A-sb-p showed killed bacteria with abnormal curling. The Schiff base cross-linked zwitterionic silicone hybrid polymer not only inactivated bacteria but also exhibited rapid and efficient bactericidal properties. For the 5% a-sb-p sample, the bactericidal rate against a 10^8 CFU/mL bacterial solution reached 90 % at 6 h and increased to 98.7 % at 12 h. For the 15% a-sb-p sample, the antibacterial rate reached 69.6 % at 12 h and achieved 99.9 % for both 5% a-sb-p and 15% a-sb-p at 24 h (Fig. 7e) that decreased by 36.5 % of 15% a-pdms, demonstrating excellent antibacterial performance. Moreover, even after 7 days of degradation, the coating maintained an antibacterial rate exceeding 99.8 % within 24 h. Following 30 days of degradation, it still achieved an antibacterial rate above 75.6 % within the same timeframe (Figs. S13–15). This demonstrated the remarkable durability and sustained antimicrobial efficacy throughout prolonged degradation periods of coating, highlighting its potential for long-term applications in environments requiring persistent antibacterial protection. Furthermore, the coating demonstrates compatibility with diverse substrates (e.g., metals, polymers, glass, fabrics) (Fig. S16), showcasing its exceptional potential for applications in marine chemical engineering, medical devices, and related industries.

3.4. Multi-modal antifouling mechanisms of hybrid coatings

The Schiff base cross-linked zwitterionic silicone polymer (A-sb-p), derived from zwitterionic silicone hybrid polymer (A-pdms), deviates from conventional single mode static antifouling strategies relying solely on disinfection or resistance mechanisms (Fig. 8a). Instead, it integrates multiple synergistic antifouling mechanisms, including silicone-mediated surface antifouling properties [8], ion hydration layer mediated antiadhesion from zwitterionic polymers [11], contact killing [12], and Schiff base enabled surface self-renewal [27] (Fig. 1a, Fig. 8b). The hybrid polymer network primarily consists of low surface energy silicone segments functionalized through grafting oligomeric sulfobetaine silane, degradable crosslinking units TPA [20], and intelligently released antibacterial CAL [17]. This design strategically combines silicone flexibility with zwitterionic hydrophilicity while leveraging the water-responsive degradation characteristics of dynamic imine bonds (Fig. 4). Within this hybrid coating system, the silicone matrix provides fouling release capabilities through its inherent low elastic modulus and low surface energy [9]. Simultaneously, the grafted zwitterionic groups establish dense hydration layer on the coating surface, thereby suppressing initial bacterial adhesion through steric hindrance and hydration effects. The introduced quaternary ammonium cations (from zwitterionic) can interact with the negatively charged bacterial membranes, compromising its structural integrity and further enhancing antibacterial efficacy [12]. This interaction induces structural destabilization via dual mechanisms, as evidenced by leakage of intracellular components from *E. coli* in immersion media of zwitterionic-modified samples (Fig. 7b). Additional modifications involving CAL and TPA increase crosslinking density between silicone segments via imine bond formation, thereby augmenting hydrophobicity. Crucially, these imine bonds undergo hydrolysis in aqueous environments, enabling controlled release of bactericidal CAL and crosslinking units TPA to facilitate surface regeneration (Fig. 4) while maintaining dynamic equilibrium and antifouling functionality. By synergistically integrating static antifouling elements (low surface energy, hydration layer) with dynamic self-renewal capabilities, this coating achieves superior antifouling performance.

4. Conclusion

In summary, based on the synergistic effect of Schiff base chemistry

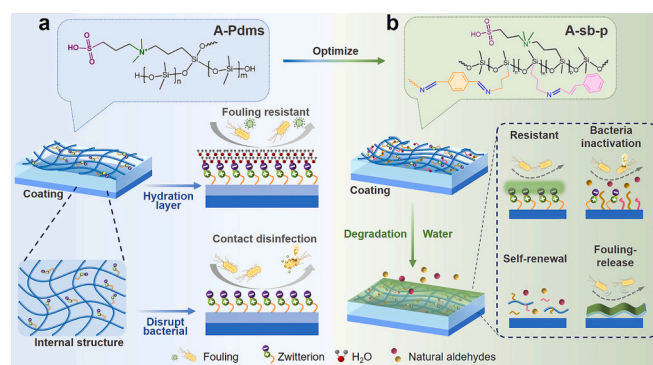


Fig. 8. (a) Bactericidal mechanism of zwitterionic sulfobetaine silane. This includes the formation of a hydrated layer on the sample surface by zwitterions to resist bacterial adhesion, and the interaction of positively charged groups in the zwitterionic structure with the bacterial cell membrane, disrupting its structural integrity. (b) Bactericidal mechanism of Schiff base cross-linked zwitterionic organosilicon hybrid coating, including inherent low surface energy of the long chain PDMS backbone promote bacterial release, a hydrated layer resisting bacterial adhesion, synergistic bactericidal action of aldehydes (from Schiff base bonds) and zwitterions, self-renewing surface properties maintaining long-term efficacy.

and zwitterionic organosilicon and through investigating the reaction mechanism between Schiff base and zwitterionic organosilicon, this study successfully developed a long-lasting, highly bactericidal, and environmentally friendly Schiff base crosslinked zwitterionic organosilicon composite coating system via molecular design and the dehydration condensation grafting of TPA and CAL onto zwitterionic organosilicon side chains. The high crosslinking capability of TPA facilitates the formation of a dense crosslinked network with the PDMS structure, significantly enhancing the stability and hydrophobicity of coating, with a contact angle greater than 99.6° . Furthermore, the incorporation of TPA improves the thermal properties of the coating. For the antibacterial performance of hybrid coating, a zwitterion mediated hydration layer formed through hydrogen bonding interactions blocks 63.4 % of bacterial invasion, while the Schiff base composite coating connected via imine bonds maintains surface stability through continuous self-renewal, achieving 99.9 % bacterial eradication efficiency and 99.7 % antiadhesion rate. Based on degradation rate calculations, the coating with a thick of 200 μm demonstrates a predicted antifouling lifespan of approximately 9.6 years, demonstrating a high efficiency and durable antibiofouling performance. This innovative composite coating uniquely integrates the static antifouling mechanism of zwitterionic organosilicon with the dynamic bactericidal mode of Schiff base chemistry, establishing a multifunctional fouling control system. It has potential for a wide range of applications in medical devices, marine infrastructure and underwater equipment.

CRedit authorship contribution statement

Yuchen Zhong: Writing – review & editing, Writing – original draft, Data curation. **Minglong Yan:** Writing – review & editing, Resources, Methodology, Funding acquisition, Data curation, Conceptualization. **Saijun Wu:** Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **An Su:** Supervision, Project administration, Conceptualization. **Jibin Pu:** Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2025.169828>.

Data availability

Data will be made available on request.

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